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Ali Mohammad Sahlodin

Assistant Professor
Department of Chemical Engineering
Process Design

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288



Educational Records

#	Degree/Major	University	City	Country	Graduation Date
1	BSc in Chemical Engineering (Process Design)	University of Tehran	Tehran	Iran	2005
2	MSc in Chemical Engineering (Process Modeling, Simulation, and Control)	Sharif University of Technology	Tehran	Iran	2007
3	PhD in Chemical Engineering (Global Optimization of Dynamic Processes)	McMaster University	Hamilton	Canada	2012

Languages

#	Language	Translation Skill	Dialogue Skill
3	English	Fluent	Fluent
3	Farsi (Persian)	Native	Native

Scientific & Industrial Experience

#	Responsibilities Titles	Name of Organization	Starting Date	Termination Date
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1	Postdoctoral Associate	Massachusetts Institute of Technology (Cambridge, MA, USA)	2013	2015
2	Research Associate	McMaster Advanced Control Consortium, McMaster University (Canada)	2013	2013
3	Senior Engineer (R&D)	Aspen Technology (MA, USA)	2016	2018

Research Interests

#	Title	Start date
1	Process Optimization under Uncertainty	
2	Process Integration and Intensification	
3	Advanced Process Control (Real-time Optimization, MPC, economic MPC)	
4	Steady-state and Dynamic Process Simulation	
5	Advanced Numerical Methods (rigorous solution of parametric dynamic systems, hybrid differential-algebraic equations)	
6	Air Pollution Modeling	

Supervised MSc Theses

#	Thesis title	By	Date
1	Modeling of continuous crystallization processes using computational fluid dynamics	MOHAMMAD ZAREI & HamidReza Noroozi Other Supervisor:	2021 Mar
2	Simulation and optimization of nitrogen production unit	LOGHMAN ENAYATZADEH	2025 Mar

Books

#	Title	Author(s)	Publisher country	publication date	version
1	Dynamic Modeling and Control of a Continuous Biopharmaceutical Manufacturing Plant Optimization of Pharmaceutical Processes	Mohammad Amin Boojari, Simone Perra, Giorgio Colombo, Matteo Grossi, Mark Nicholas Jones, Isuru Udugama, M. Nikkhah Nasab, Mohammad Fakroleslam, A.M. Sahlodin, Seyed Abbas Shojaosadati, Krist V. Gernaey, Seyed Soheil Mansouri	Switzerland		1

Journal Papers

- 1 M. Khamda, A.M. Sahlodin, N. Fallah, "A new dynamic operability index based on on-specification production for continuous pharmaceutical manufacturing under uncertainty", *Scientific Reports*, 2026 Sep vol. 0, no. , pp.1-30, 2026.
- 2 سينا ملكي, A.M. Sahlodin, N. Fallah, "Fuzzy Logic-based Gain Scheduling for PI control of a Nonlinear pH Process", *Iranian Journal of Chemical Engineering*, 2026 Aug vol. 23, no. 2, pp.91-113, 2026.
- 3 A.M. Sahlodin, R. Bozorgmehry Boozarjomehry, M. Fakhroleslam, "Dynamic Real-Time Optimization with Product Quality Considerations and Closed-loop Prediction: Application to the Tennessee Eastman Benchmark", *Scientia Iranica*, 2026 Jul vol. 33, no. , , 2026.
- 4 S. Ghanbari, A.M. Sahlodin, "Systematic inter-plant carbon integration through emission-optimal design of robust carbon dioxide reuse networks", *Journal of Cleaner Production*, 2026 Jan vol. 538, no. , pp.147297-147323, 2026.
- 5 M. Alizadeh, A.M. Sahlodin, "Monoclonal Antibodies Manufacturing in a Fed-batch Bioreactor using Economic Nonlinear Model Predictive Control with Extended Kalman Filter", *Iranian Journal of Chemistry and Chemical Engineering-International English Edition*, 2024 Nov vol. 44, no. 1, pp.298-309, 2024.
- 6 M.E. Azghandi, H. Mehdipour, A.M. Sahlodin, "Dynamic Modeling and Modification of Ternary Semicontinuous Distillation without a Middle Vessel for Improved Controllability and Energy Performance", *Chemical Engineering and Processing*, 2024 Oct vol. 205, no. 0, pp.110021-110037, 2024.
- 7 A.M. Sahlodin, Rahmat Sotudeh-gharebagh, "On Digital Technologies for Improving Chemical Engineering Education", , 2024 Aug vol. 26, no. 103, pp.67-92, 2024.
- 8 M. Alizadeh, A.M. Sahlodin, "Energy-efficient semi-continuous distillation of a ternary mixture using combined tracking-economic model predictive control", *Chemical Engineering Research and Design*, 2024 Aug vol. 209, no. 0, pp.283-298, 2024.
- 9 M. Zarei, H. Noroozi, A.M. Sahlodin, "Computational fluid dynamics simulation of a jet crystallizer for continuous crystallization of lovastatin", *Scientific Reports*, 2024 Jan vol. 14, no. 0, pp.1-16, 2024.
- 10 A.M. Sahlodin, "Optimally Safe Tank Changeover Operation Using a Smooth Optimization Formulation", *Acs Omega*, 2022 Sep vol. 0, no. 0, pp.1-16, 2022.
- 11 A.M. Fathi Moghadam, A.M. Sahlodin, Mohammad-hossein Sarrafzadeh, "A backoff approach to design of optimally flexible water networks under uncertainty", *Journal of Cleaner Production*, 2022 Aug vol. 371, no. 1, pp.1-18, 2022.
- 12 Mohammad Fakhroleslam, Ramin Bozorgmehry Boozarjomehry, A.M. Sahlodin, Grkan Sin, Seyed Soheil Mansouri, "Dynamic Simulation of Natural Gas Transmission Pipeline Systems through Autoregressive Neural Networks", *Industrial and Engineering Chemistry Research*, 2021 Jul vol. 60, no. 27, pp.9851-9859, 2021.
- 13 Changsoo Kim, Manas Shah, A.M. Sahlodin, "Design of multi-loop control systems for distillation columns: review of past and recent mathematical tools", *Chemical Product and Process Modeling*, 2021 Mar vol. 0, no. 0, pp.1-27, 2021.
- 14 A.M. Sahlodin, Benoit Chachuat, "Discretize-then-relax approach for convex/concave relaxations of the solutions of parametric ODEs", *Applied Numerical Mathematics*, 2011 Jul vol. 61, no. 7, pp.803-820, 2011.
- 15 A.M. Sahlodin, Benoit Chachuat, "Convex/concave relaxations of parametric ODEs using Taylor models", *Computers and Chemical Engineering*, 2011 May vol. 35, no. 5, pp.844-857, 2011.
- 1 Ali M. Sahlodin--- Paul Barton, "Efficient Control Discretization based on Turnpike Theory for Dynamic Optimization ", *Processes*, 2017 vol. 5, no. 4, p.85, 2017.
- 2 Ali M. Sahlodin, HAJ Watson, PI Barton, "Nonsmooth model for dynamic simulation of phase changes ", *Aiche Journal*, 2016 vol. 62, no. 9, p.3334, 2016.
- 3 Ali M. Sahlodin, PI Barton, "Optimal Campaign Continuous Manufacturing", *Industrial & Engineering Chemistry Research*, 2015 vol. 54, no. 45, p.11344, 2015.
- 4 Ali M. Sahlodin, B Chachuat, "Discretize-then-relax approach for convex/concave relaxations of the solutions of parametric ODEs", *Applied Numerical Mathematics*, 2011 vol. 61, no. 7, p.803, 2011.
- 5 Ali M. Sahlodin, B Chachuat, "Convex/concave relaxations of parametric ODEs using Taylor models", *Computers & Chemical Engineering*, 2011 vol. 35, no. 5, p.844, 2011.

- 6 M Golshan, BR BOZORGMEHRY, Ali M. Sahlodin, MR Pishvaie, " Fuzzy Real-Time Optimization of the Tennessee Eastman Challenge Process", *Iranian Journal of Chemistry and Chemical Engineering (Ijccce)*, 2011 vol. 30, no. 3, p.31, 2011.
- 7 Ali M. Sahlodin, Rahmat Sotudeh-Gharebagh, Yifang Zhu, " Modeling of dispersion near roadways based on the vehicle-induced turbulence concept", *Atmospheric Environment*, 2007 vol. 41, no. 1, p.92, 2007.

Conference Papers

- 1 M. Khamda, A.M. Sahlodin, N. Fallah, "An optimization approach to operability analysis of continuous aspirin crystallization considering feed uncertainties," *The 18th Iranian National Congress of Chemical Engineering*, Oct 2024.
- 2 M. Habibi, A.M. Sahlodin, "Hydrogen Minimization using Optimal Hydrogen Network: a Case Study," *The 18th Iranian National Congress of Chemical Engineering*, Oct 2024.
- 3 S. Ghanbari, A.M. Sahlodin, "Optimal CO2 Network Design for Reduced Greenhouse Gas Emission: A Case Study," *The 18th Iranian National Congress of Chemical Engineering*, Oct 2024.
- 4 M. Alizadeh, A.M. Sahlodin, "Shrinking-horizon Economic Model Predictive Control of a Fed-batch Reactor for Optimal MAb Production," *12th International Chemical Engineering Congress & Exhibition*, Dec 2023.
- 5 M.E. Azghandi, A.M. Sahlodin, "Optimal Side Stream Recycle Control for Semi-continuous Distillation of a Ternary Mixture," *12th International Chemical Engineering Congress & Exhibition*, Dec 2023.
- 6 M. Alizadeh, A.M. Sahlodin, Arunkumar Palanisamy, Francesco Casella, Peter Fritzson, "Application of the OpenModelica-Matlab Interface to Integrated Simulation and Successive Linearization Based Model Predictive Control," *The 15th International Modelica Conference*, Oct 2023.
- 7 A. Karraki Ebrahim Abadi, A.M. Sahlodin, "Study of the Effects of Design Parameters on the Energy Consumption and Controllability of Dividing-wall Columns," *Second National Conference on the use of Experimental and Numerical Methods in Chemical and Mineral Industries*, Feb 2023.
- 8 A.M. Fathi Moghadam, A.M. Sahlodin, Mohammad Hossein Sarrafzadeh, "A Highly Water-Efficient Corn Refinery: Design of a Water Network with a Treatment Unit," *4th Iranian Congress of Water and Wastewater Science and Engineering*, Nov 2022.
- 9 M. Mohammadkhani, A. Raisi, A.M. Sahlodin, "Simulation of hollow-fiber membrane module for gas separation using CAPE-OPEN interface," *5th National Conference on Separation Sciences and Technology*, May 2022.
- 10 M. Zarei, H. Noroozi, A.M. Sahlodin, "Modeling of continuous crystallization of lovastatin in a jet crystallizer using computational fluid dynamics," *17th Iranian National Congress of Chemical Engineering*, Nov 2021.
- 11 F. Toopchelari, A.M. Sahlodin, "Optimal Design of Flare Height Considering Safe Pollution Levels in Neighboring Areas," *11th International Chemical Engineering Congress*, Oct 2020.
- 12 M. Nikkhah Nasab, A.M. Sahlodin, "A Smooth Formulation for Maximum On-Specification Production in Campaign Continuous Manufacturing," *11th International Chemical Engineering Congress*, Oct 2020.
- 13 A.M. Sahlodin, "Process Systems Engineering: A Holistic Approach to Optimal Design and Operation of Complex Chemical Processes," *16th Iranian National Chemical Engineering Congress*, Jan 2019.
- 1 Ali M. Sahlodin, Paul I. Barton, "Optimal Campaign Continuous Manufacturing", 2015.
- 2 Ali M. Sahlodin, Paul I. Barton, "Optimal Campaign Continuous Manufacturing of Pharmaceuticals", 2015.
- 3 Ali M. Sahlodin, Paul I. Barton, "Startup Optimization of Crystallization Units in a Continuous Pharmaceutical Plant ", 2014.
- 4 Kamil A. Khan, Ali M. Sahlodin, Paul I. Barton, "Verifying Differentiability of Solutions of Nonsmooth Dynamic Systems", 2014.

- 5 Ali M. Sahlodin, Paul I. Barton, "Optimal Startup of a Continuous Pharmaceutical Process", 2014.
- 6 Benoît Chachuat, Ali M. Sahlodin, "Relaxations for the Solutions of Parametric ODEs in Global Dynamic Optimization", 2011.
- 7 Ali M. Sahlodin, Benoît Chachuat, "Tight Convex and Concave Relaxations via Taylor Models for Global Dynamic Optimization", 2011.
- 8 Ali M. Sahlodin, Benoît Chachuat, "Discretize-then-relax approach for state relaxations in global dynamic optimization", 2010.
- 9 Benoît Chachuat, Ali M. Sahlodin, "A Discretize-then-relax Approach for Convex/Concave Relaxation of the Solutions of Parametric ODEs", 2009.
- 10 Ali M. Sahlodin, Mohammad Shahrokhi, "Adaptive sliding mode control of a pH process", 2008.

Awards & Honors

#	title	Date
1	Outstanding Teaching Assistant Award, Course: Optimization in Chemical Engineering, McMaster University	2012
2	International Excellence Award, McMaster University	2011
3	Ranked 5th in the Nationwide Biotechnology Graduate Entrance Exam, Iran	2005
4	Ranked 1st among around 80 B.Sc. students in the Chem. Eng. Department, University of Tehran	2001-2005

Professional Memberships

#	Professional memberships	Explanations
1	Iranian Association of Chemical Engineering	2019
2	American Institute of Chemical Engineers	2014-2015

Executive Activities

#	Executive Activities Titles	Name of Organization / comment	Starting Date	Termination Date
1	Head of Undergraduate Process Control Laboratory	Department of Chemical Engineering, Amirkabir University of Technology	2019	
2	Head of Process Design Group	Department of Chemical Engineering, Amirkabir University of Technology	2021	2024
3	Associate Chair for Student and Cultural Affairs	Department of Chemical Engineering, Amirkabir University of Technology	2024	

Journal Referee

#	Magazine Title	Date of Cooperation & Explanations
1	Environmental Science & Technology	2020
2	Chemical Product and Process Modeling	2020
3	Scientia Iranica	2018-2019
4	Journal of Process Control	2019
5	Chemical Papers	2019
6	Optimal Control Applications and Methods	2016

Taught Courses

#	Course title	Description	Headlines	Date
1	Process Dynamics and Control	اجرای موفق فرآیندهای شیمیایی از مرحله ی راه اندازی تا خاتمه کار نیازمند وجود یک سیستم کنترلی مناسب که برآمده از شناخت درست رفتار دینامیکی		4051
2	Process modeling and simulation			4051
3	Process Dynamics and Control	اجرای موفق فرآیندهای شیمیایی از مرحله ی راه اندازی تا خاتمه کار نیازمند وجود یک سیستم کنترلی مناسب که برآمده از شناخت درست رفتار دینامیکی		4042
4	Process Safety	هدف از این درس ارایه اصول فنی در مورد ایمنی فرایند به منظور آشنایی دانشجویان با مفاهیم ایمنی و بهداشتی در صنایع شیمیایی و تجهیزات مربوطه		4042
5	Controllable Design of Process	به طور سنتی، آنچه بیشتر در مباحث طراحی فرآیند مورد تاکید می گیرد کارکرد مطلوب و بهینه آن در شرایط طراحی است. این در حالی است که هر فرآیندی		4042
6	Process Dynamics and Control			4041
7	Process modeling and simulation			4041
8	Process Design Project I			4041
9	Controllable Design of Process			4032
10	Process Safety			4032
11	Process Dynamics and Control			4032